

Unit 3: Network Layer Services, IPv4 & Routing

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Network Layer Services: Packetizing, routing, forwarding, datagram networks vs virtual-circuit networks. Network Performance: Transmission delay, propagation delay, queuing delay, processing delay, bandwidth-delay product. IPv4 A...

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Network Layer Services & Switching Models

§ 1

The **Network Layer** is responsible for host-to-host delivery of individual packets across multiple diverse network links. It encapsulates transport-layer segments into network **datagrams** at the sender, routes them across intermediate routers, and delivers them to the destination host.

THREE FUNDAMENTAL NETWORK LAYER SERVICES:

1. **Packetizing:** Encapsulating the transport-layer payload into a network-layer datagram by adding an IP header at the source, and decapsulating it at the destination.
2. **Routing:** The global decision-making process of determining the optimal, end-to-end path that packets should traverse from source to destination using routing protocols (OSPF, BGP, RIP).
3. **Forwarding:** The local action taken by each individual router of transferring a packet from an incoming interface link to the appropriate outgoing interface link based on its forwarding table.

Datagram (Connectionless) vs Virtual-Circuit (Connection-Oriented) Networks

Feature / Dimension	Datagram Approach (e.g. The Internet / IP)	Virtual-Circuit Approach (e.g. ATM, X.25)
Setup Phase	None. Packets can be injected immediately.	Mandatory connection setup phase before sending data; teardown after.
Packet Header	Contains full 32-bit (or 128-bit) Source and Destination IP addresses.	Contains only a short Virtual Circuit Identifier (VCI) label.
Routing Path	Each datagram is routed independently. Packets between same source and destination may take completely different paths.	All packets for a connection follow the exact same established physical path.

Feature / Dimension	Datagram Approach (e.g. The Internet / IP)	Virtual-Circuit Approach (e.g. ATM, X.25)
Packet Ordering	Packets may arrive out-of-order at destination; receiver must resequence.	Packets guaranteed to arrive strictly in the order sent.
Router State & Robustness	Routers maintain no per-connection state . Highly robust against router crashes (traffic automatically reroutes).	Routers maintain connection state tables. If any intermediate switch crashes, the entire connection is severed.

QUICK RECALL

- Routing = global path selection (control plane); Forwarding = local packet switching (data plane).
- The Internet uses the **Datagram model**: connectionless, independent packet routing, robust against router failure.
- Virtual-circuit uses setup/teardown phases and VCI labels.

Network Performance & Delay Modeling

§ 2

As a datagram journeys from source host to destination, it encounters four distinct delay components at every intermediate router. Understanding these delays is essential for network dimensioning and quality of service (QoS).

TOTAL NODAL DELAY EQUATION:

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

1. **Processing Delay (d_{proc}):** The time required for a router to inspect the packet header, check bit-level checksums, and look up the destination IP in the routing table (microseconds).
2. **Queuing Delay (d_{queue}):** The time a packet spends waiting in the router's transmission buffer queue before being placed on the link. Highly variable, depends on traffic congestion and queue depth.
3. **Transmission Delay (d_{trans}):** The time required to push all bits of the packet onto the physical transmission link:

$$d_{\text{trans}} = \frac{L}{R} \quad [\text{seconds}]$$

where L is packet length (bits) and R is link transmission rate / bandwidth (bps).

4. **Propagation Delay (d_{prop}):** The time required for a single bit to travel physically across the transmission medium from source to destination:

$$d_{\text{prop}} = \frac{d}{v} \quad [\text{seconds}]$$

where d is distance (meters) and v is propagation velocity of the medium ($\approx 2 \times 10^8$ m/s in copper/fiber, 3×10^8 m/s in air).

CRITICAL DISTINCTION: TRANSMISSION VS PROPAGATION DELAY:

Transmission delay (L/R) depends on packet size and transmission rate (bandwidth). It is the time to push the bits into the pipe.

Propagation delay (d/v) depends solely on physical distance and propagation velocity. It is the time for a bit to travel through the pipe.

Bandwidth-Delay Product (BDP)

The **Bandwidth-Delay Product** defines the volume of bits that can fill the transmission link end-to-end at any instant (the "pipe volume"):

$$\text{BDP} = \text{Bandwidth} \times \text{Delay} = R \times t_{\text{prop}}$$

Example: A 100 Mbps link with a 20 ms one-way propagation delay has a BDP of $(100 \times 10^6 \text{ bps}) \times (0.020 \text{ s}) = 2,000,000 \text{ bits} = 250 \text{ KB}$. The sender can emit 250 KB before the first bit reaches the receiver!

QUICK RECALL

- $d_{\text{trans}} = L/R$ (packet length / bandwidth); $d_{\text{prop}} = d/v$ (distance / speed).
- Total delay = $d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$.
- $\text{BDP} = R \times t_{\text{prop}}$ represents maximum bits in flight.

IPv4 Addressing & CIDR Subnetting

§ 3

An **IPv4 address** is a 32-bit binary identifier uniquely assigned to every host network interface card connected to the global Internet. Addresses are conventionally written in **dotted-decimal notation** (e.g. '192.168.1.1'), where each 8-bit octet is represented as a decimal number between 0 and 255.

1. Historic Classful Addressing Architecture

Class	Leading Bits	First Octet Range	Default Mask	NetID / HostID Split	Number of Networks & Hosts
Class A	'0...'	'1.0.0.0' - '126.255.255.255'	'255.0.0.0' (18)	8 bits	126 networks,
				NetID, 24 bits	$2^{24} - 2 =$
				HostID	16,777,214 hosts each
Class B	'10...'	'128.0.0.0' - '191.255.255.255'	'255.255.0.0' (16)	16 bits	16,384 networks,
				NetID, 16 bits	$2^{16} - 2 =$
				HostID	65,534 hosts each

Class	Leading Bits	First Octet Range	Default Mask	NetID / HostID Split	Number of Networks & Hosts
					2,097,152 networks, $2^8 - 2 = 254$ hosts each
Class C	<code>'110...'</code>	<code>'192.0.0.0' - '223.255.255.255'</code>	<code>'255.255.255.0' (/24)</code>	24 bits NetID, 8 bits HostID	
Class D	<code>'1110...'</code>	<code>'224.0.0.0' - '239.255.255.255'</code>	N/A	Reserved for Multicast	No NetID/HostID distinction
Class E	<code>'1111...'</code>	<code>'240.0.0.0' - '255.255.255.255'</code>	N/A	Reserved for Experimental	Future research and testing

Special Note: `'127.0.0.0/8'` is reserved for loopback testing (`127.0.0.1` = localhost).

2. Classless Inter-Domain Routing (CIDR)

Classful addressing caused massive address wastage (a company needing 500 hosts had to take a Class B with 65,534 addresses). In 1993, **CIDR (Classless Inter-Domain Routing)** abandoned fixed classes in favor of arbitrary prefix lengths specified using **slash notation (/n)**.

CIDR BLOCK MATHEMATICAL LAWS:

For an IP block specified as **A.B.C.D/n**:

1. **Prefix length (n)**: Number of 1s in the subnet mask ($0 \leq n \leq 32$).
2. **Host bits (h)**: $h = 32 - n$.
3. **Total IP addresses in block (N)**:

$$N = 2^{32-n} = 2^h$$

4. **Total assignable usable hosts (N_{usable})**:

$$N_{usable} = 2^{32-n} - 2$$

(Subtract 2: one for Network Address, one for Directed Broadcast Address).

5. **Network Address**: Bitwise AND between IP address and subnet mask.
6. **Directed Broadcast Address**: Bitwise OR between IP address and inverted mask (replace all h host bits with 1s).
7. **First Usable Host IP**: Network Address + 1.
8. **Last Usable Host IP**: Broadcast Address - 1.

WORKED NUMERICAL · VERIFIED SCRIPT PASS [8 MARKS · EXAM FAVOURITE]

Comprehensive Analysis of CIDR Block: 200.16.70.82/27

Given the host IPv4 address **200.16.70.82/27**, determine:

1. Subnet mask in dotted-decimal format
2. Network address (Subnet ID)
3. Directed broadcast address
4. Range of usable host IP addresses
5. Total number of assignable host addresses

Step 1: Subnet Mask:

Prefix length $n = 27$ means twenty-seven 1s followed by five 0s:

$$\text{Mask} = \underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{11100000}_{224} = \mathbf{255.255.255.224}$$

Step 2: Network Address:

Focus on the 4th octet: $82 = 01010010_2$. Mask 4th octet = 11100000_2 .

$$01010010_2 (82)$$

$$\text{AND } 11100000_2 (224) = 01000000_2 = 64$$

$$\text{Network Address} = \mathbf{200.16.70.64}$$

Step 3: Directed Broadcast Address:

Set all $32 - 27 = 5$ host bits in 4th octet to 1s:

$$01000000_2 \text{ OR } 00011111_2 = 01011111_2 = 64 + 31 = 95$$

$$\text{Broadcast Address} = \mathbf{200.16.70.95}$$

Step 4: Usable Host Range & Usable Count:

- First usable host: Network Address + 1 = **200.16.70.65**
- Last usable host: Broadcast Address - 1 = **200.16.70.94**
- Total usable hosts: $2^{32-27} - 2 = 2^5 - 2 = 32 - 2 =$
30 usable hosts.

Example CIDR Block: 200.16.70.82 / 27 (Subnet Mask: 255.255.255.224)
Total 32 Bits = 27 Network Prefix Bits + 5 Host Identifier Bit



Figure 3.2: CIDR Prefix Bit Partition & Host Range for 200.16.70.82/27

As required in Ramaiah SEE Examination: 27 network prefix bits leave 5 host bits, yielding 30 assignable host addresses (200.16.70.65 to 200.16.70.94).

[Interactive Visualizer Online]: Interactive IPv4 Subnet & CIDR Calculator. Explore the step-by-step visualizer on the live course notes page.

QUICK RECALL

- Class A = /8, Class B = /16, Class C = /24. Class D = Multicast.
- CIDR $/n$: $h = 32 - n$. Total IPs = 2^h ; Usable hosts = $2^h - 2$.
- Network address has all host bits = 0; Broadcast address has all host bits = 1.

VLSM Hierarchical Design & NAT Architecture

§ 4

Variable Length Subnet Masking (VLSM) enables network engineers to allocate subnets of different sizes tailored precisely to the number of hosts required in each departmental segment, completely eliminating address fragmentation.

Dept	Hosts Needed	Formula ($2^h - 2 \geq$ Needed)	Allocated Block	Subnet Mask	Usable Host IP Range
		$2^7 - 2 =$			
Dept A	120 hosts	$126 \geq 120 \Rightarrow h = 7, n = 25$	'192.168.1.0/25'	'255.255.255.128'	'192.168.1.1' - '192.168.1.126'
		$2^6 - 2 =$			
Dept B	60 hosts	$62 \geq 60 \Rightarrow h = 6, n = 26$	'192.168.1.128/26'	'255.255.255.192'	'192.168.1.129' - '192.168.1.190'
		$2^5 - 2 =$			
Dept C	30 hosts	$30 \geq 30 \Rightarrow h = 5, n = 27$	'192.168.1.192/27'	'255.255.255.224'	'192.168.1.193' - '192.168.1.222'
		$2^4 - 2 =$			
Dept D	10 hosts	$14 \geq 10 \Rightarrow h = 4, n = 28$	'192.168.1.224/28'	'255.255.255.240'	'192.168.1.225' - '192.168.1.238'

Network Address Translation (NAT)

NAT maps multiple private local IP addresses to a single globally routable public IP address, prolonging the lifespan of IPv4.

RFC 1918 PRIVATE IP ADDRESS RANGES:

- **10.0.0.0/8:** `10.0.0.0` – `10.255.255.255` (16,777,216 addresses, large enterprises).
- **172.16.0.0/12:** `172.16.0.0` – `172.31.255.255` (1,048,576 addresses, medium networks).
- **192.168.0.0/16:** `192.168.0.0` – `192.168.255.255` (65,536 addresses, home/SOHO routers).

NAPT (Network Address Port Translation): The router replaces the private IP and port with its own public IP and an ephemeral translated port (e.g. `192.168.1.5:3344` → `203.0.113.1:50001`), maintaining state in its NAT Translation Table.



Figure 3.4: Network Address Translation (NAT / NAPT) Packet Rewriting (RFC 3022)

As required in Ramaiah CIE-2 Exam Q2(a): Maps private socket (192.168.1.10:45000) to public gateway socket (203.0.113.1:50001).

QUICK RECALL

- VLSM golden rule: sort host requirements in descending order before allocating subnets.
- Private IP blocks (RFC 1918): `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`.
- NAPT maps many private hosts to one public IP using distinct transport-layer port numbers.

The IPv4 Datagram Header & Fragmentation**§ 5**

The **IPv4 Datagram Header** is a 20-to-60 byte protocol control structure prefixing every Internet packet. Because underlying physical networks possess different **Maximum Transmission Units (MTUs)**, the network layer must fragment

Structure of the IPv4 Datagram Header (32 Bits Wide)



Key Header Fields & Explanations

- **VER (4 bits)**: IP version number (always `4` for IPv4).
- **IHL (4 bits)**: Internet Header Length in 4-byte (32-bit) words. Minimum is $5 \times 4 = 20$ bytes; maximum is $15 \times 4 = 60$ bytes.
- **Total Length (16 bits)**: Total datagram length (header + payload) in bytes. Maximum theoretical size is $2^{16} - 1 = 65,535$ bytes.
- **Time to Live (TTL, 8 bits)**: Hop limit counter. Decrement by 1 at every router hop. When TTL reaches 0, the datagram is dropped and an ICMP Time Exceeded message is sent back to the source (prevents infinite routing loops).
- **Protocol (8 bits)**: Identifies the transport-layer protocol carried in payload: `1` for ICMP, `6` for TCP, `17` for UDP.
- **Flags (3 bits)**: Bit 0: Reserved (0). Bit 1: **DF (Don't Fragment)**: if 1, router cannot fragment packet; if oversized, router drops it. Bit 2: **MF (More Fragments)**: 1 means more fragments follow; 0 means this is the final fragment.
- **Fragment Offset (13 bits)**: Relative position of the fragmented data payload compared to the original unfragmented payload, measured in 8-byte (64-bit) units.

Fragment #	Payload Data (Bytes)	Total Length (with 20B Header)	Byte Range Carried	Fragment Offset (Start Byte/8)	MF Flag
Fragment 1	1480 bytes	1500 bytes	Bytes 0 - 1479	$0/8 = 0$	1 (More follow)
Fragment 2	1480 bytes	1500 bytes	Bytes 1480 - 2959	$1480/8 = 185$	1 (More follow)
Fragment 3	3980 - 2960 = 1020 B	1020 + 20 = 1040 B	Bytes 2960 - 3979	$2960/8 = 370$	0 (Final fragment)

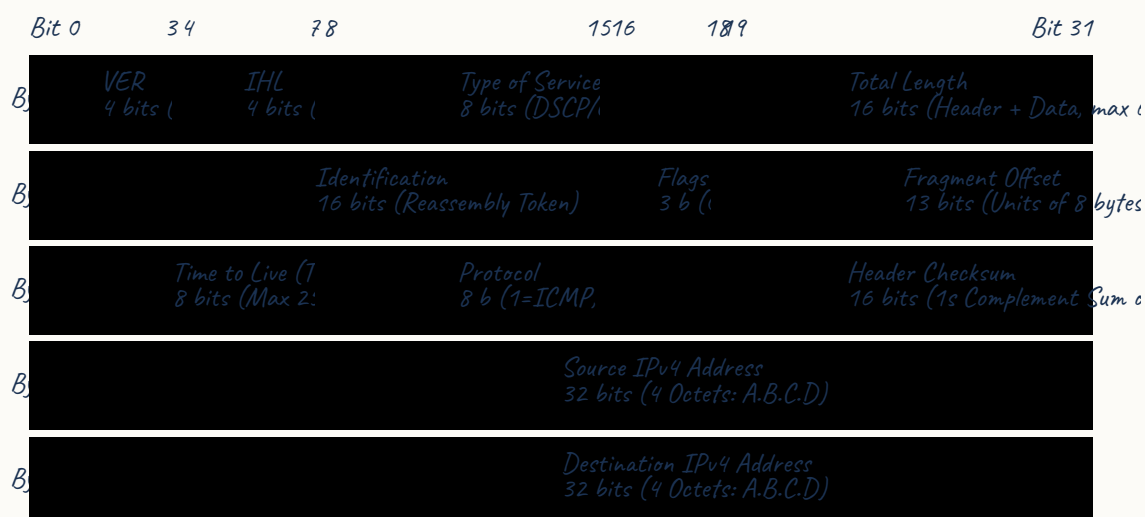


Figure 3.1: IPv4 20-Byte Base Datagram Header Structure (RFC 791)

Five 32-bit rows defining 14 standard fields (IHL=5 means $5 \times 4 = 20$ bytes; Fragment Offset in 8-byte blocks; Checksum covers header only).

QUICK RECALL

- IHL is in 4-byte units (min 5 = 20B). Total length is in bytes (max 65,535B).
- Fragment offset is strictly measured in **8-byte blocks** (Offset = Data Start Byte/8).
- MF = 1 for all fragments except the last, where MF = 0.

Internet Control Message Protocol (ICMPv4)

§ 6

IPv4 is a best-effort, connectionless datagram protocol lacking built-in error reporting or query management. **ICMPv4** operates as an companion protocol at the Network Layer to report transmission errors back to the source and provide diagnostic queries.

Category	Type	Code	Message Name	Trigger Event & Practical Application
Error-Reporting Messages	Type 3	0-15	Destination Unreachable	Router cannot find a route to the destination host, network, or port (e.g. Code 3 = Port Unreachable).
	Type 11	0	Time Exceeded	TTL reaches 0 before reaching destination. Formulates the foundation of the traceroute utility!
	Type 12	0	Parameter Problem	Header contains ambiguity or corrupted field (checksum failure or invalid option).
	Type 5	0-3	Redirection	Router informs host to update its default routing table with a more optimal local router.
Query Messages	Type 8 / 0	0	Echo Request / Echo Reply	Used by the ping utility to verify host reachability and round-trip time (RTT).
	Type 13 / 14	0	Timestamp Request / Reply	Computes clock synchronization and one-way transit delays.

RULES FOR SUPPRESSING ICMP ERROR GENERATION:

To avoid broadcast storms and infinite message loops, ICMP error messages are NEVER generated in response to:

- Another ICMP error message (no error about an error!).
- A datagram carrying a broadcast or multicast destination address.
- Any datagram fragment other than the initial fragment (offset 0).

QUICK RECALL

- ICMP runs over IP (Protocol number = 1).
- `ping` uses Type 8 (Echo Request) & Type 0 (Echo Reply).
- `traceroute` uses Type 11 (Time Exceeded, TTL=0).
- Never generate an ICMP error in response to an ICMP error or broadcast packet!

Routing Algorithms: Distance Vector & Dijkstra's Link State**§ 7**

A routing algorithm determines the optimal (least-cost) path through the network graph. The two foundational dynamic routing paradigms are **Distance Vector Routing (Bellman-Ford)** and **Link State Routing (Dijkstra)**.

1. Distance Vector Routing (Bellman-Ford Algorithm)

In Distance Vector (e.g. RIP), each router maintains a table containing its estimated distance to every destination in the network. Periodically and asynchronously, each router transmits its entire distance vector only to its **immediate neighbors**.

BELLMAN-FORD OPTIMALITY EQUATION:

$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

Where $c(x, v)$ is the cost of the direct link from router x to neighbor v , and $D_v(y)$ is neighbor v 's advertised cost to reach destination y .

COUNT-TO-INFINITY PROBLEM & REMEDIES:

Distance Vector algorithms suffer from slow convergence when links fail: "Good news travels fast, bad news travels slow." Two adjacent routers can create a routing loop by continually incrementing their path estimates through each other towards infinity.

Remedies:

- **Split Horizon:** Do not advertise a route back out the same interface through which it was learned.
- **Poison Reverse:** Advertise the route back to that neighbor with a cost of ∞ (unreachable).
- **Hold-down Timers:** Ignore route updates for a failed link for a defined period while network converges.

2. Link State Routing (Dijkstra's Algorithm)

In Link State Routing (e.g. OSPF), every router discovers its neighbors and link costs, builds a **Link State Advertisement (LSA)**, and **floods** it to every router in the entire network. Every router constructs an identical, complete map of the network topology graph and independently runs **Dijkstra's Shortest Path First (SPF)** algorithm.

Step	Permanently Visited Set N'	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	
Init	{A}	2, A	4, A	∞	C
1	{A, B} (pick B)	—	$\min(4, 2 + 1) = 3, B$	$\min(\infty, 2 + 7) = 9, B$	C
2	{A, B, C} (pick C)	—	—	9, B	D
3	{A, B, C, E} (pick E)	—	—	$\min(9, 6 + 2) = 8, E$	—
4	{A, B, C, E, D} (pick D)	—	—	—	—
5	{A, B, C, E, D, F} (pick F)	—	—	—	—

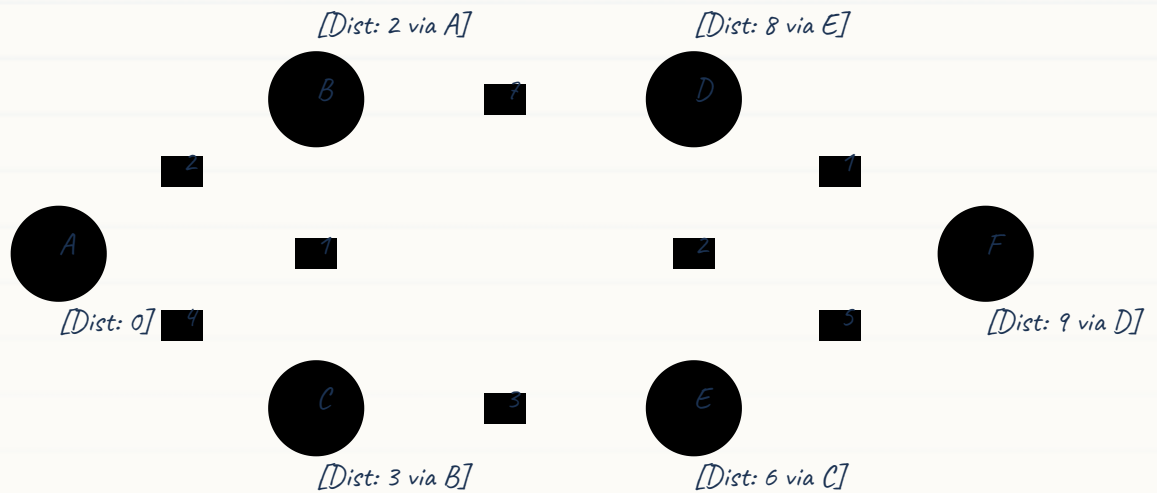


Figure 3.3: Dijkstra's Shortest Path Tree on 6-Node Network (Ramaiah CIE-2 Q3.a Solved)

Source Node A: Shortest tree paths highlighted in green (A→B: 2, A→C: 3 via B, A→E: 6 via C, A→D: 8 via E, A→F: 9 via D).

QUICK RECALL

- Distance Vector (RIP): Bellman-Ford, local neighbors, count-to-infinity problem.
- Link State (OSPF): Dijkstra SPF, global topology map via LSA flooding, $O(|V|^2)$ time.
- Dijkstra iteratively picks the unvisited node with minimum tentative cost.

One-Page Rapid Revision Sheet

REVISION

Network Delays

- $d_{trans} = L/R$ (packet length / bandwidth).
- $d_{prop} = d/v$ (distance / propagation speed).
- Total nodal delay: $d_{proc} + d_{queue} + d_{trans} + d_{prop}$.
- $BDP = R \times t_{prop}$.

CIDR & VLSM Formulas

- Prefix $/n$ has $h = 32 - n$ host bits.
- Total addresses = 2^h ; Usable hosts = $2^h - 2$.
- Network IP: AND with mask. Broadcast IP: OR with inverted mask.
- VLSM: always sort subnets by host count descending!

IPv4 Header & Frag

- IHL in 4-byte words (min 5 = 20B, max 15 = 60B).
- Fragment Offset is in **8-byte blocks**.
- MF = 1 (more fragments follow), MF = 0 (final fragment).
- TTL prevents loops; decremented at each hop.

ICMP & Routing

- ICMP Type 8/0 = Ping Echo; Type 11 = TTL Exceeded (Traceroute).
- Distance Vector: $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$.
- Dijkstra Link State: greedy min-cost expansion on flooded LSA map.

Important Examination Questions

? QUESTIONS

2-Mark Questions (Short Answer)

Practice Problem:

2M Distinguish between routing and forwarding.

Solution:

Answer: Routing is the global process of determining the end-to-end optimal path across the entire network (control plane). Forwarding is the local operation at a router transferring a packet from its input link to the appropriate output link based on its forwarding table (data plane).

Practice Problem:

2M Why is the fragment offset in an IPv4 header multiplied by 8?

Solution:

Answer: The fragment offset field is only 13 bits wide, but total datagram length can be up to 65,535 bytes (2^{16}). By measuring the offset in 8-byte units (2^3), $2^{13} \times 2^3 = 2^{16} = 65,536$, allowing the 13-bit field to span the entire 64 KB address space.

Practice Problem:

2M What is the purpose of the TTL field in the IPv4 header?

Solution:

Answer: Time-to-Live (TTL) prevents packets from circulating endlessly in routing loops. It is decremented by 1 at every router hop; when it reaches 0, the packet is discarded and an ICMP Time Exceeded message is returned to the source.

5-Mark Questions (Medium Length)

Practice Problem:

SM Compare Distance Vector Routing with Link State Routing.

Solution:

Answer:

- **Information shared:** Distance Vector shares its entire distance table only with immediate neighbors. Link State shares only link states to its immediate neighbors, but floods this to all routers in the entire network.
- **Convergence:** Distance Vector converges slowly and suffers from count-to-infinity loops. Link State converges rapidly without loops.
- **Computational complexity:** Distance Vector runs localized Bellman-Ford calculations. Link State requires every router to independently execute Dijkstra's algorithm ($O(|V|^2)$).

Practice Problem:

SM Explain Network Address Translation (NAT) and its advantages.

Solution:

Answer: NAT maps multiple private IPv4 addresses from internal networks (RFC 1918) to a single registered public IP address when communicating over the Internet. It conserves globally routable IPv4 addresses and enhances security by masking internal network topologies from outside observation.

10-Mark Questions (Long Answer / Comprehensive)

Practice Problem:

10M Draw the IPv4 datagram header format and explain the function of each field in detail.

Solution:

Answer Summary: Draw the 32-bit wide header diagram and explain all 14 fields (Version, IHL, TOS/DSCP, Total Length, Identification, Flags DF/MF, Fragment Offset, TTL, Protocol, Header Checksum, Source & Destination IP, Options) as presented in Section 5.

Solved Questions from Past CIE-2 Examination Papers

ARCHIVE

The following questions have been transcribed from the official Ramaiah CIE-2 examination paper (Term 2025) in notes/cn/practice/cn-cie-1-and-2.pdf, solved with complete intermediate steps:

Ramaiah Internal Assessment - II (2025) · Q 1(a) [7 Marks · Blooms L2, C03]

"Differentiate between IPv4 and IPv6 packet formats. Explain the role of ICMP messages in IP communication."

Part 1: Key Differences between IPv4 and IPv6 Packets:

Feature	IPv4 Header (RFC 791)	IPv6 Header (RFC 2460 / 8200)
Header Length	Variable (20 to 60 bytes, depending on Options)	Fixed 40 bytes (Base header)
Address Size	32 bits (4 octets, 2^{32} approx 4.29×10^9)	128 bits (16 octets, 2^{128} approx 3.4×10^{38})
Checksum	Included in header (recalculated at every hop)	Removed (integrity verified by L2 and L4)
Fragmentation	Handled by routers and sending host (ID, Flags, Offset)	Handled only by sending host using Extension Headers
Flow Label	Not available (only 8-bit TOS/DSCP)	20-bit dedicated Flow Label for QoS stream routing
Next Header	8-bit Protocol field	8-bit Next Header field (daisy-chains extension headers)

Part 2: Role of ICMP Messages in IP Communication:

Since IP is a best-effort, connectionless network layer protocol lacking built-in error reporting, **ICMPv4 (RFC 792)** operates alongside IP to provide:

- **Error Reporting:** Informs the sending host when a packet cannot be delivered (e.g., Type 3: Destination Unreachable, Type 11: Time to Live Exceeded during routing loops/traceroute, Type 12: Parameter Problem, Type 4: Source Quench).
- **Diagnostic Queries:** Enables reachability and latency testing via Type 8: Echo Request and Type 0: Echo Reply (ping utility), as well as timestamping for clock synchronization.

Ramaiah Internal Assessment - II (2025) · Q 1(c) [5 Marks · Blooms L2, C03]
 "What are well-known, registered, and dynamic port numbers? Explain with examples."

Port numbers are 16-bit integers (0 to 65,535) used by transport-layer protocols (TCP and UDP) to identify specific application processes on an IP host. IANA divides the port number space into three distinct ranges:

- **1. Well-Known Ports (0 to 1023):** Assigned and controlled strictly by IANA for standard universal server services. Requires superuser/root privileges to bind on UNIX systems. Examples: Port 20/21 (FTP), Port 22 (SSH), Port 25 (SMTP), Port 53 (DNS), Port 80 (HTTP), Port 443 (HTTPS).
- **2. Registered Ports (1024 to 49,151):** Not strictly assigned for universal services, but registered with IANA to avoid duplicate usage by vendor applications. Examples: Port 1433 (Microsoft SQL Server), Port 3306 (MySQL), Port 5432 (PostgreSQL), Port 8080 (HTTP Alternate / Apache Tomcat).
- **3. Dynamic / Private / Ephemeral Ports (49,152 to 65,535):** Neither controlled nor registered by IANA. Automatically and temporarily allocated by the client operating system to initiate outbound client connections. Deallocated immediately once the socket connection terminates.

Private Source (LAN)	NAT Public Source (WAN)	External Destination	Protocol
192.168.1.10:45000	203.0.113.1:50001	198.51.100.2:80 (Web)	TCP
192.168.1.11:45000	203.0.113.1:50002	198.51.100.2:80 (Web)	TCP
192.168.1.10:52140	203.0.113.1:50003	8.8.8.8:53 (DNS)	UDP

Dimension	Distance Vector Routing (e.g., RIP)	Link State Routing (e.g., OSPF)
Underlying Algorithm	Distributed Bellman-Ford algorithm	Dijkstra's Shortest Path First (SPF) algorithm
Knowledge Scope	Knows only direct neighbors and their vectors ("routing by rumor")	Every router knows full global network topology

Dimension	Distance Vector Routing (e.g., RIP)	Link State Routing (e.g., OSPF)
Updates Sent	Sends entire routing table periodically (e.g. 30s)	Sends Link-State Advertisements (LSAs) only when changes occur
Convergence Speed	Slow convergence; vulnerable to routing loops	Very fast convergence; loop-free SPF calculation
Memory & CPU	Low memory and minimal computational overhead	High memory for LSDB link database; high CPU for Dijkstra run

Ramaiah Internal Assessment - II (2025) · Q 2(c) [5 Marks · Blooms L3, C04]

"Explain TCP congestion control mechanisms: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery (TCP Tahoe vs Reno)."

TCP maintains a congestion window ($extcwnd$) and slow-start threshold ($extssthresh$) to prevent network collapse:

- **1. Slow Start:** Begins with $extcwnd = 1extMSS$. For every ACK received, $extcwnd$ increments by 1 MSS, causing exponential growth ($extcwnd$ doubles each RTT: 1 to 4 to 8). Continues until $extcwnd \geq extssthresh$.
- **2. Congestion Avoidance:** Once $extcwnd \geq extssthresh$, growth switches to linear (Additive Increase: $+1extMSS$ per RTT) to probe channel capacity conservatively.
- **3. Fast Retransmit:** When sender receives 3 duplicate ACKs for the same segment, it assumes the packet was lost and retransmits immediately without waiting for the retransmission timeout (RTO).
- **4. Fast Recovery (Tahoe vs Reno):**
 - **TCP Tahoe:** On packet loss (timeout OR 3 dup ACKs), sets $extssthresh = extcwnd/2$ and resets $extcwnd = 1extMSS$, falling back into Slow Start.
 - **TCP Reno:** On 3 duplicate ACKs, sets $extssthresh = extcwnd/2$ and sets $extcwnd = extssthresh + 3extMSS$ (Fast Recovery), avoiding the severe penalty of dropping back to 1 MSS. On timeout, Reno still drops to $extcwnd = 1extMSS$.

Step	N' (Visited)	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	$D(E), p(E)$
0 (Init)	{A}	2, A	4, A	∞	∞
1	{A, B}	2, A	3, B	9, B	∞
2	{A, B, C}	2, A	3, B	9, B	6, C
3	{A, B, C, E}	2, A	3, B	8, E	6, C
4	{A, B, C, E, D}	2, A	3, B	8, E	6, C
5 (Done)	{A, B, C, E, D, F}	2, A	3, B	8, E	6, C

Ramaiah Internal Assessment - II (2025) · Q 3(b) [10 Marks · Blooms L2, C03]
"Discuss the transition mechanisms from IPv4 to IPv6: Dual Stack, Tunneling, and Header Translation."

Because the global Internet cannot be upgraded from IPv4 to IPv6 overnight, the IETF standardized three fundamental co-existence strategies:

1. 1. Dual Stack (RFC 4213):

- Hosts and routers run both IPv4 and IPv6 protocol stacks simultaneously on the same physical interfaces.
- When an application initiates communication, DNS queries return both an A record (IPv4) and an AAAA record (IPv6). If IPv6 is available, the client prioritizes the IPv6 stack; otherwise, it falls back seamlessly to IPv4.
- Advantage: Cleanest migration mechanism; no protocol translation overhead.

2. 2. Tunneling (e.g., 6to4, 6in4, Teredo):

- Used when two IPv6 nodes or networks must communicate across an intervening IPv4-only transit network.
- The ingress border router encapsulates the entire IPv6 datagram as payload inside an IPv4 datagram (Protocol 41). The packet traverses the IPv4 network as normal traffic.
- The egress border router decapsulates the packet, strips the IPv4 header, and forwards the native IPv6 packet to the destination.

3. 3. Header Translation (NAT-PT / SIIT / NAT64):

- Required when an IPv6-only host must communicate directly with an IPv4-only host that cannot be upgraded.
- The translator device strips the IPv6 header and synthesizes an equivalent IPv4 header (mapping 128-bit addresses to 32-bit addresses and adjusting total length, TTL/Hop Limit, and checksums).
- Limitation: Information loss can occur because IPv6 lacks certain IPv4 header fields, and applications with embedded IP addresses (like FTP) require Application Layer Gateways (ALGs).

Practice Problems with Step-by-Step Solutions

PRACTICE

Test your understanding with these exam-level numericals. Click each collapsible card to reveal the complete verified solution:

PROBLEM 1 · CIDR ADDRESSING

An organization is assigned the CIDR block **198.51.100.160/26**. Determine: (a) Subnet Mask in dotted decimal, (b) Network Address, (c) Broadcast Address, and (d) Range of usable host IP addresses.

Practice Problem:

Show Step-by-Step Solution

Solution:**Step 1: Subnet Mask**

Prefix length $n = 26$. First 3 octets are all 1s (24 bits). 4th octet has 2 ones ($11000000_2 = 192_{10}$).

$$\text{extSubnetMask} = \mathbf{255.255.255.192}$$

Step 2: Network Address

Bitwise AND on 4th octet: $160_{10} = 10100000_2$.

$$10100000_2 \text{extAND} 11000000_2 = 10000000_2 = 128_{10}.$$

$$\text{extNetworkAddress} = \mathbf{198.51.100.128}$$

Step 3: Broadcast Address

Host bits $h = 32 - 26 = 6$. Total addresses in block $= 2^6 = 64$.

$$\text{Broadcast} = 128 + 64 - 1 = 191.$$

$$\text{extBroadcastAddress} = \mathbf{198.51.100.191}$$

Step 4: Usable Host Range

$$\text{extUsableHosts} = \mathbf{198.51.100.129 \text{extto} 198.51.100.190}$$

PROBLEM 2 · VLSM ALLOCATION

An enterprise base block is **192.168.10.0/24**. Allocate subnets for three departments: Dept A (60 hosts), Dept B (28 hosts), and Dept C (12 hosts). Specify network address, mask, and host range for each.

Practice Problem:

Show Step-by-Step Solution

Solution:

Always allocate starting from the largest requirement:

- **Dept A (60 hosts):** Needs $60 + 2 = 62$ addresses. Nearest power of 2 is $2^6 = 64$ ($h = 6$). Subnet prefix: $32 - 6 = /26$ (Mask 255.255.255.192).
 - Network: 192.168.10.0/26, Range: 192.168.10.1 - 192.168.10.62, Broadcast: 192.168.10.63.
- **Dept B (28 hosts):** Needs $28 + 2 = 30$ addresses. Nearest power of 2 is $2^5 = 32$ ($h = 5$). Subnet prefix: $32 - 5 = /27$ (Mask 255.255.255.224).
 - Starts at next free boundary: 192.168.10.64/27, Range: 192.168.10.65 - 192.168.10.94, Broadcast: 192.168.10.95.
- **Dept C (12 hosts):** Needs $12 + 2 = 14$ addresses. Nearest power of 2 is $2^4 = 16$ ($h = 4$). Subnet prefix: $32 - 4 = /28$ (Mask 255.255.255.240).
 - Starts at next free boundary: 192.168.10.96/28, Range: 192.168.10.97 - 192.168.10.110, Broadcast: 192.168.10.111.

PROBLEM 3 · IP FRAGMENTATION

An IPv4 packet with Total Length = 4000 bytes (20 bytes header, 3980 bytes payload) arrives at an outgoing link with MTU = 1500 bytes. Calculate the parameters (Total Length, Data Size, Offset, DF, and MF) for all fragments.

Practice Problem:

Show Step-by-Step Solution

Solution:

Max payload per fragment = $1500 - 20 = 1480$ bytes. Notice 1480 is divisible by 8 ($1480/8 = 185$).

- **Fragment 1:**

- Data: 1480 bytes (bytes 0 to 1479).
- Total Length: $1480 + 20 = 1500$ extbytes.
- Fragment Offset: $0/8 = 0$.
- Flags: $extDF = 0, extMF = 1$.

- **Fragment 2:**

- Data: 1480 bytes (bytes 1480 to 2959).
- Total Length: $1480 + 20 = 1500$ extbytes.
- Fragment Offset: $1480/8 = 185$.
- Flags: $extDF = 0, extMF = 1$.

- **Fragment 3:**

- Remaining Data: $3980 - (1480 + 1480) = 1020$ bytes (bytes 2960 to 3979).
- Total Length: $1020 + 20 = 1040$ extbytes.
- Fragment Offset: $2960/8 = 370$.
- Flags: $extDF = 0, extMF = 0$ (Last fragment).

PROBLEM 4 · TRANSMISSION VS PROPAGATION DELAY

A 1500-byte packet is sent over a 2500 km fiber-optic cable ($v = 2 \times 10^8 \text{ m/s}$) at transmission rate $R = 1 \text{ Gbps}$. Calculate d_{trans} , d_{prop} , and determine whether the link is latency-dominated or bandwidth-dominated.

Practice Problem:

Show Step-by-Step Solution

Solution:**1. Transmission Delay:**

$$L = 1500 \times 8 = 12,000 \text{ bits}$$

$$d_{\text{trans}} = \frac{L}{R} = \frac{12,000 \text{ bits}}{10^9 \text{ bps}} = 1.2 \times 10^{-5} \text{ s} = \mathbf{12 \text{ } \mu\text{s}}$$

2. Propagation Delay:

$$d = 2,500 \text{ km} = 2.5 \times 10^6 \text{ m}$$

$$d_{\text{prop}} = \frac{d}{v} = \frac{2.5 \times 10^6}{2 \times 10^8} = 0.0125 \text{ s} = \mathbf{12.5 \text{ ms}} = \mathbf{12,500 \text{ } \mu\text{s}}$$

3. Latency Regime:

$$d_{\text{prop}} / d_{\text{trans}} = 12,500 / 12 \approx \mathbf{1041.7}$$

Since propagation delay exceeds transmission delay by over 1000 times, this long-haul link is heavily **propagation/latency-dominated**.

PROBLEM 5 · DIJKSTRA SHORTEST PATH

In a diamond network with nodes W, X, Y, Z and bidirectional edge costs $W-X=3, W-Y=5, X-Y=1, X-Z=6, Y-Z=2$, determine the shortest path and cost from W to Z using Dijkstra's algorithm.

Practice Problem:

Show Step-by-Step Solution

Solution:**Execution Trace from Source W :**

1. Init: $D(X) = 3, D(Y) = 5, D(Z) = \infty$. Min is X ($D = 3$).

Permanent: $\{W, X\}$.

2. From X : update $D(Y) = \min(5, 3 + 1) = 4$ via X . Update

$D(Z) = \min(\infty, 3 + 6) = 9$ via X . Min is Y ($D = 4$).

Permanent: $\{W, X, Y\}$.

3. From Y : update $D(Z) = \min(9, 4 + 2) = 6$ via Y . Min is Z ($D = 6$).

Permanent: $\{W, X, Y, Z\}$.

extShortestPath : $W \rightarrow X \rightarrow Y \rightarrow Z$ extwithCost = $3 + 1 + 2 =$

PROBLEM 6 · ICMP ERROR CODE ANALYSIS

A host runs traceroute to a target server. (a) What ICMP error message is generated by intermediate routers? (b) What ICMP error message is generated by the destination host when the probe reaches it?

Practice Problem:

Show Step-by-Step Solution

Solution:**(a) Intermediate Routers:**

The host sends UDP probe packets with incrementally increasing TTL values ($TTL = 1, 2, 3 \dots$). When router hop k decrements TTL to 0, it drops the packet and transmits back to the source:

extICMPType11, Code0 : TimetoLiveExceededinTra

(b) Destination Host:

The packet reaches the final destination with $TTL > 0$. Because the traceroute client targets an intentionally unused high UDP port (e.g., 33434+), the destination host's transport layer cannot deliver the datagram and returns:

extICMPType3, Code3 : DestinationUnreachable(Por